Tiered Bibliography

Quantum Mechanics & Quantum Chemistry Programme

Quantum Mechanics — Reading Lists by Learning Tier

Tier 1	High School — Divulgation & First Introductions
Tier 2	Beginning Undergraduate — Standard Textbooks
Tier 3	Advanced Undergraduate — Rigorous Texts & Historical Papers
Tier 4	Master's — Advanced Textbooks, Monographs & Research Papers
Tier 5	PhD — Research Monographs, Specialised Topics & Primary Literature
Cat. A–G	Full Alphabetical Catalogue (all tiers)

★★★ *Essential* ★★ *Strongly recommended* ★ *Useful supplement*

[N] *Popular/narrative* ***[T]*** *Textbook* ***[P]*** *Primary paper* ***[M]*** *Monograph* ***[R]*** *Review*

Contents

How to Use This Bibliography

This bibliography accompanies the **Pedagogical Support Material** and the **Course Syllabus** of the QM & QC Programme. It is purely quantum-mechanical in scope, covering quantum mechanics, quantum chemistry, and their mathematical foundations. Reading lists are stratified by the same five tiers used for content delivery and assessment. The tiers are cumulative: Tier k reading is assumed as background for Tier $k+1$.

Tier	Label	PSM track	Primary reading mode
1	HS	Undergrad	Popular books cover-to-cover before the course
2	BegUG	Undergrad	One textbook chapter per lecture; do exercises
3	AdvUG	Undergrad	Textbook + primary paper abstracts and introductions
4	MSc	Graduate	Primary papers + advanced textbooks as main reading
5	PhD	Grad + Research	Monographs + full primary literature + arXiv

1 Tier 1 — Divulgation, Popular Science & First Introductions

Tier 1 — High School (HS): Popular Science & Divulgation Books

Goal: Build vivid physical intuition and historical awareness. No mathematical prerequisites beyond arithmetic and basic algebra. Read these books *before* any formal course to establish the mental pictures that make the formalism meaningful. They answer "What is quantum mechanics and why does it matter?" not "How does it work mathematically?"

1.1 Divulgation & Popular Science

Feynman, R. P.
QED: The Strange Theory of Light and Matter
Princeton University Press, 1985.
★★★ [N]

The finest popular account of quantum amplitudes and the path-integral picture. Feynman explains probability amplitudes as "arrows" that rotate and combine — exactly the pre-formal picture for Module I.2 Lectures L01–L02. Essential pre-reading for anyone entering the programme.

Feynman, R. P., Leighton, R. B. & Sands, M.
The Feynman Lectures on Physics, Vol. III: Quantum Mechanics
Addison-Wesley, 1965. Freely at `feynmanlectures.caltech.edu`
★★★ [N/T]

Chapters 1–4 (interference, amplitude, identical particles) are accessible at Tier 1 with no calculus. These chapters provide the pre-formal motivation for Module I.2 Lectures L01–L04 and L09. A free, authoritative resource.

Cox, B. & Forshaw, J.
The Quantum Universe: Everything That Can Happen Does Happen
Da Capo Press, 2011.
★★ [N]

A carefully accurate popular account covering superposition, entanglement, and the exclusion principle. Recommended as complementary Tier 1 reading alongside Feynman's QED.

Kumar, M.
Quantum: Einstein, Bohr, and the Great Debate about the Nature of Reality
Norton, 2008.
★★ [N]

Historical narrative of the foundational debates: Copenhagen interpretation, EPR paradox, Schrödinger's cat. Essential context for understanding why measurement theory (Module I.2 Lecture L07, Module I.3 Lecture L10) was so controversial.

Gamow, G.
Thirty Years That Shook Physics: The Story of Quantum Theory
Dover, 1966.
★★ **[N]**

A charming historical account by a participant. Covers Planck's quantum, Bohr's atom, de Broglie, Schrödinger, Heisenberg, and Dirac with light but accurate physics. Excellent for historical context before Module I.1.

Penrose, R.
The Emperor's New Mind, Chapters 5–6
Oxford University Press, 1989.
★ **[N]**

Philosophical context for quantum mechanics, wave functions, and measurement. Chapters 5–6 are accessible at Tier 1. Provides useful philosophical grounding before Module I.3 Lectures L05–L07.

Al-Khalili, J.
Quantum: A Guide for the Perplexed
Weidenfeld & Nicolson, 2003.
★★ **[N]**

Beautifully illustrated popular account. Covers wave–particle duality, uncertainty, entanglement, and quantum technology with minimal formalism. Effective visual introduction for Tier 1 students.

Rae, A. I. M.
Quantum Physics: Illusion or Reality? (2nd ed.)
Cambridge University Press, 2004.
★★ **[N]**

Focused on the conceptual and philosophical issues: the measurement problem, hidden variables, many worlds, and Bell's theorem. Excellent bridge between popular science and the philosophical content of Module I.2 Lecture L07.

Hey, T. & Walters, P.
The New Quantum Universe (revised ed.)

Cambridge University Press, 2003.
★ **[N]**

Visually rich survey of quantum mechanics and its applications (lasers, transistors, MRI, cryptography). Good for motivating why quantum mechanics matters before the formalism of Series I.

Rovelli, C.
Helgoland: Making Sense of the Quantum Revolution
Riverhead Books, 2021.
★ **[N]**

A philosophical-historical account advocating the relational interpretation. Stimulating for students curious about interpretations of quantum mechanics; provides a counterpoint to Copenhagen before Module I.3 Lecture L10.

2 Tier 2 — Standard Undergraduate Textbooks

Tier 2 — Beginning Undergraduate (BegUG): Standard Textbooks

Goal: Develop working fluency in quantum mechanics notation and methods. Assumes first-year calculus and linear algebra. Work through one chapter per lecture, solving end-of-chapter exercises alongside SPSU problem sets from the PSM.

2.1 Core Quantum Mechanics Textbooks

Griffiths, D. J. & Schroeter, D. F.
Introduction to Quantum Mechanics (3rd ed.)
Cambridge University Press, 2018.
★★★ [T]

The standard first course globally. Develops wave mechanics before Dirac notation. Chapters 1–3 (wavefunction, Schrödinger equation, formalism) parallel Module I.3 Lectures L02–L06; Chapter 3 introduces Hilbert space and bra-ket, paralleling Module I.2 Lectures L01–L06. Superb problem sets at the SPSU level.

Shankar, R.
Principles of Quantum Mechanics (2nd ed.)
Springer, 1994.
★★★ [T]

Begins with a self-contained linear algebra review (Chapter 1) — ideal for bridging Tier 1 to Tier 2. More algebraic than Griffiths; a natural bridge to Sakurai. Chapter 1 pairs with Module I.1; Chapters 4–9 support Module I.2 Lectures L03–L08.

Townsend, J. S.
A Modern Approach to Quantum Mechanics (2nd ed.)
University Science Books, 2012.
★★ [T]

Starts from spin-$\frac{1}{2}$ and Dirac notation from chapter one (Sakurai-style). Excellent for building bra-ket fluency early. Module I.2 uses this as a parallel text for Lectures L01–L04 and L10.

Binney, J. & Skinner, D.
The Physics of Quantum Mechanics
Oxford University Press, 2014. Freely at `www-thphys.physics.ox.ac.uk`
★★ [T]

Uses bra-ket from Chapter 1. Excellent treatment of measurement and operators. Good parallel text for Module I.2 Lectures L03–L07. Free access is a strong advantage.

French, A. P. & Taylor, E. F.
An Introduction to Quantum Physics
Norton, 1978.
★★ [**T**]

An MIT undergraduate text emphasising conceptual clarity and connection to experiment. Chapters on wave packets, uncertainty, and the hydrogen atom are particularly well motivated. Good supplementary reading for Module I.3 Lectures L02, L03, L11.

McMahon, D.
Quantum Mechanics Demystified (2nd ed.)
McGraw-Hill, 2005.
★ [**T**]

A problem-solving companion with worked examples at every step. Not a substitute for a proper textbook but useful alongside Griffiths for students who need more step-by-step worked solutions. Supports SPSU practice.

Phillips, A. C.
Introduction to Quantum Mechanics
Wiley, 2003.
★ [**T**]

A concise undergraduate text well suited to a one-semester course. Good on the physical motivation for wave mechanics and the connection between classical and quantum descriptions. Supports Module I.3 Lectures L05–L07.

2.2 Mathematics Support

Strang, G.
Introduction to Linear Algebra (5th ed.)
Wellesley–Cambridge Press, 2016.
★★★ [**T**]

The prerequisite linear algebra text. Chapters 5–6 (eigenvalues, orthogonality) are directly needed for quantum mechanics. Work through before or alongside Module I.1.

Boas, M. L.
Mathematical Methods in the Physical Sciences (3rd ed.)
Wiley, 2006.
★★ **[T]**

The standard mathematical methods text for physics undergraduates. Chapter 3 (linear algebra), Chapter 6 (complex analysis), and Chapter 7 (Fourier series) are directly used in Module I.2 Lectures L02, L05, L06.

Byron, F. W. & Fuller, R. W.
Mathematics of Classical and Quantum Physics
Dover, 1992.
★★ **[T]**

Covers Hilbert spaces, operators, distributions, and Green's functions at a level accessible to advanced Tier 2 students. Useful mathematical supplement for Module I.2 Lectures L06 and L12.

2.3 Quantum Chemistry Introduction

Jensen, F.
Introduction to Computational Chemistry (3rd ed.)
Wiley, 2017.
★★ **[T]**

Accessible entry to computational quantum chemistry. Chapters 1–3 provide the Tier 2 background for Series III and should be read alongside Module III.1.

Levine, I. N.
Quantum Chemistry (7th ed.)
Pearson, 2014.
★★ **[T]**

The standard undergraduate quantum chemistry textbook. Clear development of the Schrödinger equation, variational method, and perturbation theory before moving to molecular orbital theory. Supports Module III.1 and early Module III.2.

Atkins, P. W. & Friedman, R.
Molecular Quantum Mechanics (5th ed.)
Oxford University Press, 2011.
★★ **[T]**

Bridges undergraduate physical chemistry and quantum mechanics. Excellent treatment of angular momentum, hydrogen atom, and molecular orbital theory. Supports Module II.3 and Module III.2 Lectures L02–L08.

3 Tier 3 — Rigorous Undergraduate, Bridge Texts & Historical Papers

Tier 3 — Advanced Undergraduate (AdvUG): Rigorous Texts, Bridge Books & First Primary Papers

Goal: Understand internal consistency, proofs, and the geometric structure of the formalism. Begin reading primary papers at the abstract and introduction level. Assumes mathematical analysis and linear algebra at a proof-writing level.

3.1 Core Texts

Sakurai, J. J. & Napolitano, J.
Modern Quantum Mechanics (3rd ed.)
Cambridge University Press, 2020.
★★★ **[T]**

The primary text for Module I.2. Begins with bra-ket notation and the Stern–Gerlach experiment. Chapters 1–3 (state vectors, dynamics, angular momentum) are the core of Series I. The Sakurai "operational" track is one of the two tracks in the Module I.2 dual pedagogy.

Dirac, P. A. M.
The Principles of Quantum Mechanics (4th ed.)
Oxford University Press, 1958 (reprinted 1981).
★★★ **[T/P]**

The original source for bra-ket notation. Chapter I ("The Principle of Superposition") and Chapter II ("Dynamical Variables and Observables") must be read alongside Module I.2. This is the "Dirac abstract" track of the dual pedagogy. A primary source that functions as a textbook.

Messiah, A.
Quantum Mechanics (2 vols., translated from French)
North-Holland, 1961–1962 (Dover reprint 2014).
★★★ **[T]**

One of the great classical treatises. Volume I covers wave mechanics, the Schrödinger equation, and the hydrogen atom with extraordinary clarity and rigour. Volume II covers angular momentum, perturbation theory, scattering, and identical particles. The appendices on Hilbert space and group theory are among the best introductions available. Supports all of Series I and Module II.3.

Gasiorowicz, S.
Quantum Physics (3rd ed.)
Wiley, 2003.

★★ **[T]**

A rigorous and physically well-motivated undergraduate text. Excellent on the connection between classical and quantum physics, with a thorough treatment of perturbation theory and scattering. Supports Module I.3 Lectures L09–L11.

Isham, C. J.
Lectures on Quantum Theory: Mathematical and Structural Foundations
Imperial College Press, 1995.

★★ **[T]**

Bridges undergraduate and graduate levels. Rigorous treatment of Hilbert spaces, operators, and measurement without full functional analysis. Ideal for Tier 3 students preparing for Tier 4; directly supports Module I.2 Lectures L11–L12.

Hannabuss, K.
An Introduction to Quantum Theory
Oxford University Press, 1997.

★★ **[T]**

Mathematically precise undergraduate text. Good on spectral theory, uncertainty, and density operators without requiring measure theory. Supports Module I.2 Lectures L04, L07, and L11.

Auletta, G., Fortunato, M. & Parisi, G.
Quantum Mechanics
Cambridge University Press, 2009.

★★ **[T]**

A comprehensive modern undergraduate-to-graduate text covering both formalism and interpretation. Notable for its treatment of quantum information, entanglement, and decoherence at an accessible level. Supports Module I.2 Lectures L09–L11 and Module I.3 Lecture L10.

Peres, A.
Quantum Theory: Concepts and Methods
Kluwer, 1993 (Springer reprint 2002).

★★ **[T]**

A masterful rigorous treatment with a strong emphasis on the conceptual foundations, Bell inequalities, and the measurement problem. The treatment of EPR and entanglement is particularly excellent. Supports Module I.2 Lectures L07, L09, and L11 from a foundational perspective.

3.2 Historical Primary Papers — First Encounters

Heisenberg, W.
Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen
Z. Phys. **33**, 879–893 (1925). English translation in van der Waerden (ed.), *Sources of Quantum Mechanics*, Dover, 1967.
★★ **[P]**

The paper that founded matrix mechanics. Heisenberg replaces classical observables with transition amplitudes and derives the commutation relations from a "reinterpretation" of kinematics. The historical origin of the operator formalism introduced in Module I.1 Lecture L01.

Born, M. & Jordan, P.
Zur Quantenmechanik
Z. Phys. **34**, 858–888 (1925). Translation in van der Waerden (ed.), 1967.
★★ **[P]**

The first systematic formulation of matrix mechanics: canonical commutation relations $[\hat{q}, \hat{p}] = i\hbar$, energy matrix, equations of motion. The direct historical predecessor of Module I.3 Lecture L07.

Born, M.
Zur Quantenmechanik der Stoßvorgänge
Z. Phys. **37**, 863–867 (1926). Translation in Wheeler & Zurek, *Quantum Theory and Measurement*, Princeton, 1983.
★★★ **[P]**

Two pages that changed physics: the Born rule $P = |\psi|^2$. Must be read by all Tier 3 students. Directly relevant to Module I.2 Lecture L01 and the WRQ prompt "How did Born's interpretation change the meaning of the wavefunction?"

Schrödinger, E.
Quantisierung als Eigenwertproblem (Parts I–IV)
Ann. Phys. **79**, 361 (1926); **79**, 489 (1926); **80**, 437 (1926); **81**, 109 (1926). Translations in *Wave Mechanics*, Pergamon, 1928.
★★ **[P]**

The four papers establishing wave mechanics: the wave equation, its eigenvalue problem, and the hydrogen atom solution. Part II contains Schrödinger's proof that matrix and wave mechanics are equivalent. Historical foundations for Module I.3 Lectures L05 and L11.

Heisenberg, W.
Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik

Z. Phys. **43**, 172–198 (1927). Translation in Wheeler & Zurek, 1983.

★★ **[P]**

The uncertainty principle paper. Heisenberg derives the position-momentum uncertainty relation via thought experiments (microscope). Historical foundation for Module I.2 Lecture L07 and Module I.3 Lecture L03.

Wigner, E. P.
On unitary representations of the inhomogeneous Lorentz group
Ann. Math. **40**, 149–204 (1939).

★ **[P]**

Original source for Wigner's theorem: symmetries preserve transition probabilities, hence are implemented by unitary or antiunitary operators. Abstract accessible at Tier 3; full proof requires Tier 5. Relevant to Module I.2 Lectures L01 and L03.

Einstein, A., Podolsky, B. & Rosen, N.
Can quantum-mechanical description of physical reality be considered complete?
Phys. Rev. **47**, 777–780 (1935).

★★ **[P]**

The EPR paper: the argument that quantum mechanics is incomplete via entangled states. Three pages; fully accessible at Tier 3. Essential historical reading for Module I.2 Lecture L09 (tensor products and entanglement).

Bell, J. S.
On the Einstein Podolsky Rosen paradox
Physics **1**, 195–200 (1964). Reprinted in Bell, *Speakable and Unspeakable in Quantum Mechanics*, Cambridge, 1987.

★★★ **[P]**

Bell's theorem: no local hidden-variable theory can reproduce all predictions of quantum mechanics. Six pages; the most important theorem in the foundations of quantum mechanics. Essential Tier 3 reading; directly relevant to Module I.2 Lecture L07 and the ORQ on operational axioms.

4 Tier 4 — Advanced Textbooks, Monographs & Research Papers

Tier 4 — Master's (MSc): Advanced Textbooks, Specialised Topics & Key Research Papers

Goal: Command the full formalism; reproduce key derivations independently; situate results in the broader research literature. Primary papers become the main reading; textbooks serve as reference. Functional analysis and group theory background assumed. Use alongside CPSG, SPJG, CPJG, and WRQ artifact types from the PSM.

4.1 Advanced Quantum Mechanics Textbooks

Weinberg, S.
Lectures on Quantum Mechanics (2nd ed.)
Cambridge University Press, 2015.
★★★ [T]

Covers foundations (postulates, measurement, entanglement), symmetries, and perturbation theory at graduate level. Chapter 3 ("Mathematical Tools") is the cleanest graduate treatment of bra-ket including rigged Hilbert spaces. Supports Module I.2 Lectures L11–L12 and Module I.3 Lectures L01–L09.

Galindo, A. & Pascual, P.
Quantum Mechanics (2 vols.)
Springer, 1990–1991.
★★★ [T]

The most mathematically complete standard graduate text. Volume I develops Hilbert space rigorously with spectral theory and self-adjoint operators; Volume II covers scattering, relativistic QM, and second quantisation. Recommended for Tier 4 students aiming at mathematical physics. Supports all of Series I and II.

Ballentine, L. E.
Quantum Mechanics: A Modern Development (2nd ed.)
World Scientific, 2014.
★★★ [T]

Statistical (ensemble) interpretation; detailed treatment of measurement, density operators, and the relationship between classical and quantum mechanics. Excellent for Module I.2 Lectures L07 and L11, and Module I.3 Lecture L10.

Cohen-Tannoudji, C., Diu, B. & Laloë, F.
Quantum Mechanics (2 vols., new ed.)
Wiley, 2019.
★★ **[T]**

Encyclopaedic reference. The "Complements" to each chapter are especially valuable; they cover applications, generalisations, and extensions that go beyond the main text. Supports Modules II.3 (angular momentum), II.6 (spectroscopy), and I.3.

Schwabl, F.
Advanced Quantum Mechanics (4th ed.)
Springer, 2008.
★★ **[T]**

Second quantisation, many-body theory, relativistic quantum mechanics, and path integrals at the MSc level. A compact bridge to Tier 5 monographs. Supports Modules II.2 and II.4.

Le Bellac, M.
Quantum Physics
Cambridge University Press, 2006.
★★ **[T]**

Modern graduate text emphasising quantum information, open systems, and quantum optics alongside traditional topics. Excellent on density matrices, decoherence, and quantum channels. Supports Module I.2 Lectures L08–L12 and Module I.3 Lecture L10.

4.2 Symmetry, Group Theory & Angular Momentum

Wigner, E. P.
Group Theory and Its Application to the Quantum Mechanics of Atomic Spectra
Academic Press, 1959 (translated from the 1931 German edition).
★★★ **[T/M]**

The foundational text on symmetry in quantum mechanics. Wigner develops representation theory, the rotation group, and selection rules. Chapters 1–15 support Module II.3. A primary source that remains the most authoritative treatment.

Georgi, H.
Lie Algebras in Particle Physics (2nd ed.)
Westview Press, 1999.
★★ **[T]**

The standard physics graduate text on Lie algebras and representation theory. $SU(2)$, $SU(3)$, and general semisimple algebras. Supports Module II.3 Lectures L01–L07 and the algebraic content of Module I.1.

Biedenharn, L. C. & Louck, J. D.
Angular Momentum in Quantum Physics (2 vols.)
Addison-Wesley, 1981. (Encyclopedia of Mathematics and Its Applications.)
★★ **[M]**

The definitive encyclopaedic reference for angular momentum, Clebsch–Gordan coefficients, Wigner symbols, and the Wigner–Eckart theorem. Supports Module II.3 Lectures L05–L10 at graduate depth.

4.3 Many-Body Theory

Fetter, A. L. & Walecka, J. D.
Quantum Theory of Many-Particle Systems
McGraw-Hill, 1971 (Dover reprint 2003).
★★★ **[T]**

The standard graduate text for Series II Modules II.4–II.5. Second quantisation, Green's functions, Feynman diagrams, and linear response developed systematically. Chapter 1 pairs with Module II.4 Lecture L01; Chapters 7–9 with Module II.5.

Bruus, H. & Flensberg, K.
Introduction to Many-Body Quantum Theory in Condensed Matter Physics
Oxford University Press, 2004. Freely available online.
★★★ **[T]**

More accessible than Fetter–Walecka. Excellent on second quantisation, diagrammatics, and Green's functions for condensed matter applications. Primary reading for Module II.4 Lectures L01–L07.

Stefanucci, G. & van Leeuwen, R.
Nonequilibrium Many-Body Theory of Quantum Systems
Cambridge University Press, 2013.
★★★ **[T]**

The definitive text for Module II.5 (NEGF). Derives the Keldysh formalism rigorously from first principles; self-energies, Kadanoff–Baym equations, and steady-state transport developed in full. Chapters 4–8 are primary reading for Module II.5 Lectures L02–L09.

Mahan, G. D.
Many-Particle Physics (3rd ed.)
Kluwer/Plenum, 2000.
★★ [T]

A comprehensive graduate text on many-body methods in condensed matter. Green's functions, self-energy, diagrammatic perturbation theory, and transport. Supports Modules II.4–II.6 as a reference text alongside Fetter–Walecka.

4.4 Quantum Statistical Mechanics

Huang, K.
Statistical Mechanics (2nd ed.)
Wiley, 1987.
★★ [T]

The standard graduate statistical mechanics text. Chapters on quantum statistical mechanics, Bose–Einstein condensation, and the Ising model support Module II.7.

Sachdev, S.
Quantum Phase Transitions (2nd ed.)
Cambridge University Press, 2011.
★★ [T]

The authoritative text on quantum phase transitions, quantum criticality, and non-Fermi liquids. Supports Module II.7 Lectures L09–L12.

4.5 Quantum Chemistry (Series III)

Szabo, A. & Ostlund, N. S.
Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory
McGraw-Hill, 1989 (Dover reprint 1996).
★★★ [T]

The standard entry-level graduate text for Series III. Hartree–Fock, CI, MP2, and the rudiments of DFT developed concisely with full derivations. Chapters 1–3 support Module III.1; Chapters 4–6 support Module III.2 Lectures L05–L11.

Helgaker, T., Jørgensen, P. & Olsen, J.
Molecular Electronic-Structure Theory
Wiley, 2000.
★★★ [M]

The definitive graduate reference for Modules III.2–III.3. 900 pages covering basis sets, HF, MP, CC, MCSCF, and response theory. Use as the reference text alongside Module III.2 from Lecture L07 onward.

Koch, W. & Holthausen, M. C.
A Chemist's Guide to Density Functional Theory (2nd ed.)
Wiley-VCH, 2001.
★★ **[T]**

Accessible DFT text for Tier 4 chemistry students. Good companion for Module III.3 Lectures L01–L05.

Martin, R. M.
Electronic Structure: Basic Theory and Practical Methods
Cambridge University Press, 2004.
★★ **[T]**

DFT from first principles: plane waves, pseudopotentials, PAW, and band structure. Complements the molecular orbital perspective of Module III.2 with the solid-state perspective of Module III.2 Lecture L16 and Module III.3.

4.6 Path Integrals

Feynman, R. P. & Hibbs, A. R.
Quantum Mechanics and Path Integrals (emended ed., Styer)
Dover, 2010 (original McGraw-Hill, 1965).
★★★ **[T]**

The original path-integral textbook. Develops the Feynman sum-over-histories from physical principles. Essential reading for Module II.2 Lectures L01–L04.

Zinn-Justin, J.
Quantum Field Theory and Critical Phenomena (4th ed.)
Oxford University Press, 2002.
★★ **[M]**

The standard graduate reference for path integrals in field theory. Parts I–II (functional integrals, Gaussian integrals, perturbation theory) support Module II.2 Lectures L03–L09.

4.7 Key Tier 4 Research Papers (QM & QC)

Hohenberg, P. & Kohn, W.
Inhomogeneous electron gas
Phys. Rev. **136**, B864–B871 (1964).
★★★ **[P]**

Proves the two Hohenberg–Kohn theorems establishing DFT. Primary reading for Module III.3 Lecture L01.

Kohn, W. & Sham, L. J.
Self-consistent equations including exchange and correlation effects
Phys. Rev. **140**, A1133–A1138 (1965).
★★★ **[P]**

The Kohn–Sham DFT paper: maps the interacting problem onto a non-interacting Kohn–Sham system. Primary reading for Module III.3 Lecture L03.

Perdew, J. P. & Zunger, A.
Self-interaction correction to density-functional approximations for many-electron systems
Phys. Rev. B **23**, 5048 (1981).
★★ **[P]**

Foundational DFT paper on self-interaction error and LDA. Key reading for Module III.3 Lecture L04 (XC functionals).

Becke, A. D.
Density-functional thermochemistry. III. The role of exact exchange
J. Chem. Phys. **98**, 5648–5652 (1993).
★★ **[P]**

The paper introducing the B3LYP hybrid functional — the most widely used functional in quantum chemistry. Essential context for Module III.3 Lecture L04.

Perdew, J. P., Burke, K. & Ernzerhof, M.
Generalized gradient approximation made simple
Phys. Rev. Lett. **77**, 3865–3868 (1996).
★★ **[P]**

The PBE functional paper — the standard GGA functional in solid-state calculations. Essential alongside B3LYP for Module III.3 Lecture L04.

Kadanoff, L. P. & Baym, G.
Quantum Statistical Mechanics
Benjamin, 1962 (Perseus reprint 1989).
★★★ **[M]**

Original derivation of the Kadanoff–Baym equations for the two-time Green's function. Primary reading for Module II.5 Lecture L05.

Berry, M. V.
Quantal phase factors accompanying adiabatic changes
Proc. R. Soc. Lond. A **392**, 45–57 (1984).
★★★ **[P]**

Berry's original paper on geometric phases. Directly relevant to Module II.2 Lecture L12 (Berry phase from path integrals) and Module I.3 Lecture L06 (non-commuting time evolution).

Pulay, P.
Ab initio calculation of force constants and equilibrium geometries in polyatomic molecules
Mol. Phys. **17**, 197–204 (1969).
★★ **[P]**

Introduces analytic gradient theory ("Pulay forces") in quantum chemistry — the foundation of modern geometry optimisation. Supports Module III.1 Lecture L03 (Feynman–Hellmann theorem and force calculations).

Runge, E. & Gross, E. K. U.
Density-functional theory for time-dependent systems
Phys. Rev. Lett. **52**, 997–1000 (1984).
★★ **[P]**

The Runge–Gross theorem establishing time-dependent DFT. Background reading for Module III.3 and for the linear response content of Module II.6.

Kubo, R.
Statistical-mechanical theory of irreversible processes. I.
J. Phys. Soc. Japan **12**, 570–586 (1957).
★★★ **[P]**

The Kubo formula for linear response. Primary reading for Module II.6 Lecture L03.

5 Tier 5 — Research Monographs, Specialised Topics & Primary Literature

Tier 5 — PhD: Research Monographs, Specialised Topics & Full Primary Literature

Goal: Achieve research-level mastery; read primary literature directly; identify open problems; connect the programme to current research frontiers. Primary papers and monographs are the main reading. Use alongside CPJG and ORQ artifact types from the PSM.

5.1 Functional Analysis & Mathematical Foundations of QM

Reed, M. & Simon, B.
Methods of Modern Mathematical Physics (4 vols.)
Academic Press, 1972–1979.
Vol. I: *Functional Analysis.* Vol. II: *Fourier Analysis, Self-Adjointness.* Vol. III: *Scattering Theory.* Vol. IV: *Analysis of Operators.*
★★★ [M]

The mathematical foundation of quantum mechanics at the PhD level. Vol. I: Hilbert spaces, operators, spectral theory in full rigour. Vol. II: self-adjoint extensions, deficiency indices, the Gel'fand triple. Vol. IV: spectral theory of Schrödinger operators. Essential for Module I.2 Lecture L12 and Module I.3 Lecture L04. Every PhD student in mathematical physics should work through Vol. I–II.

Rudin, W.
Functional Analysis (2nd ed.)
McGraw-Hill, 1991.
★★★ [M]

Banach and Hilbert spaces, the spectral theorem for bounded operators, distributions. Chapter 12 (unbounded operators) is directly relevant to the self-adjoint extension problem for $\hat{p}$. Supports Module I.2 Lecture L12 at PhD level.

Blank, J., Exner, P. & Havlíček, M.
Hilbert Space Operators in Quantum Physics (2nd ed.)
Springer, 2008.
★★★ [M]

The most physics-oriented rigorous mathematical treatment. Spectral measures, rigged Hilbert spaces, quantum dynamics, and quantum fields. Directly supports Module I.2 Lectures L11–L12 and Module I.3 Lectures L04–L08.

Gel'fand, I. M. & Vilenkin, N. Ya.
Generalized Functions, Vol. 4: Applications of Harmonic Analysis
Academic Press, 1964.
★★ [M]

Original source for the rigged Hilbert space (Gel'fand triple) framework $\Phi \subset \mathcal{H} \subset \Phi'$. Chapter 1 is the foundational reference for Module I.2 Lecture L12.

Thirring, W.
A Course in Mathematical Physics (4 vols.)
Springer, 1978–1983.
Vol. 3: *Quantum Mechanics of Atoms and Molecules.* Vol. 4: *Quantum Mechanics of Large Systems.*
★★ [M]

Mathematically rigorous treatment of quantum mechanics using C^*-algebras and operator algebras. Vol. 3 covers the spectral theory of atomic Hamiltonians; Vol. 4 treats quantum statistical mechanics and stability of matter. Essential background for algebraic approaches to Module II.7.

Lieb, E. H. & Loss, M.
Analysis (2nd ed.)
American Mathematical Society, 2001.
★★ [M]

Covers the functional analysis and real analysis needed for rigorous quantum mechanics: L^p spaces, Sobolev spaces, variational methods, and sharp inequalities. The Lieb–Thirring inequality and stability-of-matter results are treated here. PhD-level background for Modules III.1 and III.3.

5.2 Foundations of Quantum Mechanics

von Neumann, J.
Mathematical Foundations of Quantum Mechanics
Princeton University Press, 1955 (translated from German, 1932).
★★★ [M]

The foundational monograph. Chapters I–II: Hilbert space and operators. Chapter III: quantum statistics and measurement. Chapter VI: the measurement problem. Chapters III–VI require Tier 5 preparation. The historical and conceptual foundation for Module I.2 Lecture L12 and Module I.3 Lecture L10.

Haag, R.
Local Quantum Physics: Fields, Particles, Algebras (2nd ed.)
Springer, 1996.

★★ [M]

Algebraic QFT (AQFT): observables as C^*-algebras, superselection sectors, Tomita–Takesaki modular theory. The deepest mathematical treatment of locality and the measurement problem. Background for Series II Module II.4 at research level.

Kraus, K.
States, Effects, and Operations: Fundamental Notions of Quantum Theory
Springer Lecture Notes in Physics **190**, 1983.
★★ [M]

Kraus operators, CPTP maps, and the operational (instrument) formulation of quantum mechanics. The mathematical background for Module I.2 Lecture L11 and Module I.3 Lecture L10 at PhD depth.

Busch, P., Lahti, P. & Mittelstaedt, P.
The Quantum Theory of Measurement (2nd ed.)
Springer Lecture Notes in Physics **m2**, 1996.
★★ [M]

Rigorous axiomatic treatment of quantum measurement: POVMs, instruments, repeatability, and the Lüders rule. The mathematical foundation for Module I.2 Lecture L07 and L11 at PhD level.

5.3 Advanced Many-Body Theory & Quantum Field Theory

Negele, J. W. & Orland, H.
Quantum Many-Particle Systems
Addison-Wesley, 1988 (Perseus reprint 1998).
★★★ [M]

Path integrals and Green's functions at research level: functional methods, coherent state path integrals, finite-temperature diagrammatics. Supports Modules II.2, II.4, and II.5.

Peskin, M. E. & Schroeder, D. V.
An Introduction to Quantum Field Theory
Addison-Wesley, 1995.
★★ [T]

The standard QFT graduate text. Parts I–II (canonical quantisation, Feynman rules, one-loop divergences) provide the QFT background for Module II.4 Lecture L12 (beyond mean field, connection to QFT) and Module II.2 Lecture L09.

Altland, A. & Simons, B. D.
Condensed Matter Field Theory (2nd ed.)
Cambridge University Press, 2010.
★★★ **[T]**

The modern graduate text bridging quantum field theory and condensed matter. Path integrals, functional methods, the renormalisation group, and topology in condensed matter. Primary reading for Module II.2 and a key reference for Modules II.4–II.5.

Abrikosov, A. A., Gorkov, L. P. & Dzyaloshinski, I. E.
Methods of Quantum Field Theory in Statistical Physics
Dover, 1975 (translated from the 1962 Russian edition).
★★ **[M]**

The classic Soviet text on diagrammatic techniques for many-body systems. Green's functions, self-energy, phonon–electron interactions, and superconductivity. Supports Modules II.4 and II.5 at research level.

5.4 Quantum Information & Open Quantum Systems

Nielsen, M. A. & Chuang, I. L.
Quantum Computation and Quantum Information
Cambridge University Press, 2000.
★★★ **[M]**

The definitive reference for quantum information and quantum computing. Chapters 2 (linear algebra and density operators), 8 (quantum noise), and 9–10 (error correction) are directly relevant to Module I.2 Lectures L09–L11 and Module I.3 Lecture L10 at research depth.

Breuer, H.-P. & Petruccione, F.
The Theory of Open Quantum Systems
Oxford University Press, 2002.
★★★ **[M]**

The standard monograph on open quantum systems: Lindblad master equations, quantum Brownian motion, decoherence, and quantum trajectories. Primary reference for Module I.3 Lecture L10 and Module II.5 Lecture L11.

Carmichael, H. J.
An Open Systems Approach to Quantum Optics
Springer Lecture Notes in Physics **m18**, 1993.
★★ [M]

Quantum trajectories and the stochastic Schrödinger equation. A key tool for numerical simulation of open quantum systems. Supports Module I.3 Lecture L10 and Module II.6 at research depth.

5.5 Advanced Quantum Chemistry (Series III Research Level)

Martin, R. M., Reining, L. & Ceperley, D.
Interacting Electrons: Theory and Computational Approaches
Cambridge University Press, 2016.
★★★ [M]

The definitive modern text connecting DFT, many-body perturbation theory (GW, BSE), and quantum Monte Carlo. All of Series III covered at research depth.

Parr, R. G. & Yang, W.
Density-Functional Theory of Atoms and Molecules
Oxford University Press, 1989.
★★★ [M]

The foundational DFT monograph by its principal architects. Develops HK theorems, KS equations, the local density approximation, and chemical reactivity indices (Fukui functions, hardness/softness). Primary reference for Module III.3 Lectures L01–L05.

Dreizler, R. M. & Gross, E. K. U.
Density Functional Theory: An Approach to the Quantum Many-Body Problem
Springer, 1990.
★★ [M]

Rigorous treatment of DFT from a many-body physics perspective. Covers HK theorems, KS equations, LDA, and TDDFT. Supports Module III.3 at research level.

5.6 Essential PhD-Level Primary Papers

Stone, M. H.
On one-parameter unitary groups in Hilbert space
Ann. Math. **33**, 643–648 (1932).
★★★ [P]

Stone's theorem: the mathematical basis for $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$. Primary reading for Module I.3 Lecture L04 at PhD level.

Gleason, A. M.
Measures on the closed subspaces of a Hilbert space
J. Math. Mech. **6**, 885–893 (1957).
★★★ **[P]**

Derives the Born rule from measure theory on the lattice of closed subspaces: the deepest axiomatic justification of $P = |\langle a|\psi\rangle|^2$. Essential for Module I.2 Lecture L07 at PhD level.

Lindblad, G.
On the generators of quantum dynamical semigroups
Commun. Math. Phys. **48**, 119–130 (1976).
★★★ **[P]**

Derives the most general Markovian master equation (Lindblad equation) from complete positivity and trace preservation. Primary reading for Module I.3 Lecture L10 and Module II.5 Lecture L11.

Gorini, V., Kossakowski, A. & Sudarshan, E. C. G.
Completely positive dynamical semigroups of N-level systems
J. Math. Phys. **17**, 821–825 (1976).
★★★ **[P]**

Simultaneous with Lindblad (1976): independently derives the GKSL master equation for finite-dimensional systems. Together with Lindblad's paper, establishes the complete mathematical framework for open quantum dynamics.

Aspect, A., Grangier, P. & Roger, G.
Experimental tests of Bell's inequalities using time-varying analyzers
Phys. Rev. Lett. **49**, 1804–1807 (1982).
★★★ **[P]**

The definitive experimental test of Bell's theorem. Closes the locality loophole and firmly establishes quantum nonlocality. Essential reading alongside Bell (1964) for Module I.2 Lecture L07 and L09 at PhD level.

Heisenberg, W.
Zur Theorie des Ferromagnetismus
Z. Phys. **49**, 619–636 (1928).

★★ [P]

The Heisenberg exchange interaction: exchange integral from antisymmetry of electron wavefunctions. Historical foundation for Module II.4 Lecture L11 (quantum magnetism).

Bardeen, J., Cooper, L. N. & Schrieffer, J. R.
Theory of superconductivity
Phys. Rev. **108**, 1175–1204 (1957).
★★★ [P]

The BCS theory: Cooper pairs, the BCS gap equation, and the Bogoliubov–Valatin transformation. Primary reading for Module II.4 Lecture L09 (BCS superconductivity).

Keldysh, L. V.
Diagram technique for nonequilibrium processes
Zh. Eksp. Teor. Fiz. **47**, 1515 (1964); Sov. Phys. JETP **20**, 1018 (1965).
★★★ [P]

The Keldysh contour and closed-time-path formalism for non-equilibrium Green's functions. The foundational paper for Module II.5 Lectures L02–L04.

Meir, Y. & Wingreen, N. S.
Landauer formula for the current through an interacting electron region
Phys. Rev. Lett. **68**, 2512–2515 (1992).
★★ [P]

Derives the Meir–Wingreen formula for current through a correlated region. The central result of Module II.5 Lecture L08 (steady-state transport).

Aharonov, Y. & Bohm, D.
Significance of electromagnetic potentials in the quantum theory
Phys. Rev. **115**, 485–491 (1959).
★★★ [P]

The Aharonov–Bohm effect: the vector potential is physically significant even in regions of zero field. The quantum mechanics of gauge fields. Primary reading for Module II.2 Lecture L10.

Zurek, W. H.
Pointer basis of quantum apparatus: Into what mixture does the wave packet collapse?
Phys. Rev. D **24**, 1516–1525 (1981).

★★ [P]

Introduces the concept of environment-induced superselection ("einselection") and pointer states. The foundational paper of decoherence theory. Primary reading for Module I.3 Lecture L10 (open systems and measurement).

Dirac, P. A. M.
The quantum theory of the emission and absorption of radiation
Proc. R. Soc. Lond. A **114**, 243–265 (1927).
★★ [P]

The first paper on quantum electrodynamics: quantisation of the electromagnetic field, creation/annihilation operators, and spontaneous emission. Historical foundation for Module II.4 and Module II.6 Lecture L01.

Wigner, E. P.
The unreasonable effectiveness of mathematics in the natural sciences
Commun. Pure Appl. Math. **13**, 1–14 (1960).
★ [P]

A philosophical essay on the mysterious alignment between abstract mathematics and physical law. Stimulating reading for PhD students reflecting on the foundations of the Hilbert space formalism.

5.7 Review Articles & Survey Monographs

Leggett, A. J.
Quantum Liquids: Bose Condensation and Cooper Pairing in Condensed-Matter Systems
Oxford University Press, 2006. [M]
★★★

Research-level survey of BEC and BCS from a unified perspective. Supports Module II.4 Lectures L08–L09 and connects to Module II.7.

Zurek, W. H. [R]
Decoherence, einselection, and the quantum origins of the classical
Rev. Mod. Phys. **75**, 715–775 (2003).
★★★

The comprehensive review of decoherence theory. Covers environment-induced superselection, pointer states, quantum Darwinism, and the emergence of classicality. Primary review reading for Module I.3 Lecture L10 and Module II.5 Lecture L11.

Koch, J. et al. [R]
Charge-insensitive qubit design derived from the Cooper pair box
Phys. Rev. A **76**, 042319 (2007).
★★

The transmon qubit paper. A key paper in the application of Module I.2's formalism (two-level systems, Josephson junctions) to superconducting quantum computing. Excellent PhD-level example of the Jaynes–Cummings Hamiltonian in practice.

Rotter, S. & Fan, S. [R]
Optical cavities during the age of quantum information: Cavity QED and non-Hermitian physics
Science **358**, 1444–1447 (2017).
★

Modern review connecting cavity QED (Module II.6 Lecture L12) to non-Hermitian physics and exceptional points. Represents a current research frontier.

6 Full Alphabetical Catalogue

Complete alphabetically ordered catalogue grouped by category. Each entry carries tier badge(s), type tag, and star rating. **Categories:** A Popular/Narrative B Undergraduate Textbooks C Graduate Textbooks D Monographs E Primary Papers F Reviews G Historical Collections

A Popular Science & Divulgation

[A1] T1 Al-Khalili, J. *Quantum: A Guide for the Perplexed.* Weidenfeld & Nicolson, 2003. [N] ★★

[A2] T1 Cox, B. & Forshaw, J. *The Quantum Universe.* Da Capo Press, 2011. [N] ★★

[A3] T1 Feynman, R. P. *QED: The Strange Theory of Light and Matter.* Princeton University Press, 1985. [N] ★★★

[A4] T1 Feynman, R. P., Leighton, R. B. & Sands, M. *The Feynman Lectures on Physics, Vol. III.* Addison-Wesley, 1965. [N/T] ★★★

[A5] T1 Gamow, G. *Thirty Years That Shook Physics.* Dover, 1966. [N] ★★

[A6] T1 Hey, T. & Walters, P. *The New Quantum Universe.* Cambridge University Press, 2003. [N] ★

[A7] T1 Kumar, M. *Quantum: Einstein, Bohr, and the Great Debate.* Norton, 2008. [N] ★★

[A8] T1 Penrose, R. *The Emperor's New Mind.* Oxford University Press, 1989. [N] ★

[A9] T1 Rae, A. I. M. *Quantum Physics: Illusion or Reality?* Cambridge University Press, 2004. [N] ★★

[A10] T1 Rovelli, C. *Helgoland.* Riverhead Books, 2021. [N] ★

B Undergraduate Textbooks (Tiers 2–3)

[B1] T2 Atkins, P. W. & Friedman, R. *Molecular Quantum Mechanics* (5th ed.). Oxford University Press, 2011. [T]

[B2] T2 Binney, J. & Skinner, D. *The Physics of Quantum Mechanics.* Oxford University Press, 2014. [T] ★★

[B3] T2 Boas, M. L. *Mathematical Methods in the Physical Sciences* (3rd ed.). Wiley, 2006. [T] ★★

[B4] T2 Byron, F. W. & Fuller, R. W. *Mathematics of Classical and Quantum Physics.* Dover, 1992. [T] ★★

[B5] T3 Dirac, P. A. M. *The Principles of Quantum Mechanics* (4th ed.). Oxford University Press, 1958. [T/P] ★★★

[B6] T2 French, A. P. & Taylor, E. F. *An Introduction to Quantum Physics.* Norton, 1978. [T] ★★

[B7] T3 Gasiorowicz, S. *Quantum Physics* (3rd ed.). Wiley, 2003. [T] ★★

[B8] **T2** Griffiths, D. J. & Schroeter, D. F. *Introduction to Quantum Mechanics* (3rd ed.). Cambridge University Press, 2018. [T] ★★★

[B9] **T3** Hannabuss, K. *An Introduction to Quantum Theory.* Oxford University Press, 1997. [T] ★★

[B10] **T3** Isham, C. J. *Lectures on Quantum Theory.* Imperial College Press, 1995. [T] ★★

[B11] **T2** Jensen, F. *Introduction to Computational Chemistry* (3rd ed.). Wiley, 2017. [T] ★★

[B12] **T2** Levine, I. N. *Quantum Chemistry* (7th ed.). Pearson, 2014. [T] ★★

[B13] **T2** McMahon, D. *Quantum Mechanics Demystified* (2nd ed.). McGraw-Hill, 2005. [T] ★

[B14] **T3** Messiah, A. *Quantum Mechanics* (2 vols.). North-Holland, 1961–1962 (Dover, 2014). [T] ★★★

[B15] **T3** Peres, A. *Quantum Theory: Concepts and Methods.* Kluwer, 1993. [T] ★★

[B16] **T2** Phillips, A. C. *Introduction to Quantum Mechanics.* Wiley, 2003. [T] ★

[B17] **T3** Auletta, G., Fortunato, M. & Parisi, G. *Quantum Mechanics.* Cambridge University Press, 2009. [T] ★★

[B18] **T3** Sakurai, J. J. & Napolitano, J. *Modern Quantum Mechanics* (3rd ed.). Cambridge University Press, 2020. [T] ★★★

[B19] **T2** Shankar, R. *Principles of Quantum Mechanics* (2nd ed.). Springer, 1994. [T] ★★★

[B20] **T2** Strang, G. *Introduction to Linear Algebra* (5th ed.). Wellesley–Cambridge Press, 2016. [T] ★★★

[B21] **T2** Townsend, J. S. *A Modern Approach to Quantum Mechanics* (2nd ed.). University Science Books, 2012. [T] ★★

C Graduate Textbooks (Tiers 4–5)

[C1] **T5** Abrikosov, A. A., Gorkov, L. P. & Dzyaloshinski, I. E. *Methods of Quantum Field Theory in Statistical Physics.* Dover, 1975. [M] ★★

[C2] **T5** Altland, A. & Simons, B. D. *Condensed Matter Field Theory* (2nd ed.). Cambridge University Press, 2010. [T] ★★★

[C3] **T4** Ballentine, L. E. *Quantum Mechanics: A Modern Development* (2nd ed.). World Scientific, 2014. [T] ★★★

[C4] **T5** Blank, J., Exner, P. & Havlíček, M. *Hilbert Space Operators in Quantum Physics* (2nd ed.). Springer, 2008. [M] ★★★

[C5] **T4** Bruus, H. & Flensberg, K. *Introduction to Many-Body Quantum Theory.* Oxford University Press, 2004. [T] ★★★

[C6] **T4** Cohen-Tannoudji, C., Diu, B. & Laloë, F. *Quantum Mechanics* (2 vols., new ed.). Wiley, 2019. [T] ★★

[C7] **T4** Feynman, R. P. & Hibbs, A. R. *Quantum Mechanics and Path Integrals* (emended ed.). Dover, 2010. [T] ★★★

[C8] **T4** Fetter, A. L. & Walecka, J. D. *Quantum Theory of Many-Particle Systems.* Dover, 2003. [T] ★★★

[C9] **T4** Galindo, A. & Pascual, P. *Quantum Mechanics* (2 vols.). Springer, 1990–1991. [T] ★★★

[C10] **T4** Georgi, H. *Lie Algebras in Particle Physics* (2nd ed.). Westview Press, 1999. [T] ★★

[C11] **T4** Helgaker, T., Jørgensen, P. & Olsen, J. *Molecular Electronic-Structure Theory.* Wiley, 2000. [M] ★★★

[C12] **T4** Huang, K. *Statistical Mechanics* (2nd ed.). Wiley, 1987. [T] ★★

[C13] **T4** Koch, W. & Holthausen, M. C. *A Chemist's Guide to Density Functional Theory* (2nd ed.). Wiley-VCH, 2001. [T] ★★

[C14] **T4** Le Bellac, M. *Quantum Physics.* Cambridge University Press, 2006. [T] ★★

[C15] **T4** Mahan, G. D. *Many-Particle Physics* (3rd ed.). Kluwer, 2000. [T] ★★

[C16] **T4** Martin, R. M. *Electronic Structure.* Cambridge University Press, 2004. [T] ★★

[C17] **T4** Martin, R. M., Reining, L. & Ceperley, D. *Interacting Electrons.* Cambridge University Press, 2016. [M] ★★★

[C18] **T5** Peskin, M. E. & Schroeder, D. V. *An Introduction to Quantum Field Theory.* Addison-Wesley, 1995. [T] ★★

[C19] **T5** Rudin, W. *Functional Analysis* (2nd ed.). McGraw-Hill, 1991. [M] ★★★

[C20] **T4** Sachdev, S. *Quantum Phase Transitions* (2nd ed.). Cambridge University Press, 2011. [T] ★★

[C21] **T4** Schwabl, F. *Advanced Quantum Mechanics* (4th ed.). Springer, 2008. [T] ★★

[C22] **T4** Stefanucci, G. & van Leeuwen, R. *Nonequilibrium Many-Body Theory of Quantum Systems.* Cambridge University Press, 2013. [T] ★★★

[C23] **T4** Szabo, A. & Ostlund, N. S. *Modern Quantum Chemistry.* Dover, 1996. [T] ★★★

[C24] **T4** Weinberg, S. *Lectures on Quantum Mechanics* (2nd ed.). Cambridge University Press, 2015. [T] ★★★

[C25] **T4** Wigner, E. P. *Group Theory and Its Application to the Quantum Mechanics of Atomic Spectra.* Academic Press, 1959. [T/M] ★★★

[C26] **T4** Zinn-Justin, J. *Quantum Field Theory and Critical Phenomena* (4th ed.). Oxford University Press, 2002. [M] ★★

D Research Monographs (Tier 5)

[D1] T4 T5 Biedenharn, L. C. & Louck, J. D. *Angular Momentum in Quantum Physics* (2 vols.). Addison-Wesley, 1981. [M] ★★

[D2] T5 Breuer, H.-P. & Petruccione, F. *The Theory of Open Quantum Systems.* Oxford University Press, 2002. [M] ★★★

[D3] T5 Busch, P., Lahti, P. & Mittelstaedt, P. *The Quantum Theory of Measurement* (2nd ed.). Springer, 1996. [M] ★★

[D4] T5 Carmichael, H. J. *An Open Systems Approach to Quantum Optics.* Springer, 1993. [M] ★★

[D5] T5 Dreizler, R. M. & Gross, E. K. U. *Density Functional Theory.* Springer, 1990. [M] ★★

[D6] T5 Gel'fand, I. M. & Vilenkin, N. Ya. *Generalized Functions, Vol. 4.* Academic Press, 1964. [M] ★★

[D7] T5 Haag, R. *Local Quantum Physics* (2nd ed.). Springer, 1996. [M] ★★

[D8] T5 Kadanoff, L. P. & Baym, G. *Quantum Statistical Mechanics.* Perseus, 1989. [M] ★★★

[D9] T5 Kraus, K. *States, Effects, and Operations.* Springer, 1983. [M] ★★

[D10] T5 Leggett, A. J. *Quantum Liquids.* Oxford University Press, 2006. [M] ★★★

[D11] T5 Lieb, E. H. & Loss, M. *Analysis* (2nd ed.). AMS, 2001. [M] ★★

[D12] T5 Negele, J. W. & Orland, H. *Quantum Many-Particle Systems.* Perseus, 1998. [M] ★★★

[D13] T5 Nielsen, M. A. & Chuang, I. L. *Quantum Computation and Quantum Information.* Cambridge University Press, 2000. [M] ★★★

[D14] T5 Parr, R. G. & Yang, W. *Density-Functional Theory of Atoms and Molecules.* Oxford University Press, 1989. [M] ★★★

[D15] T5 Reed, M. & Simon, B. *Methods of Modern Mathematical Physics* (4 vols.). Academic Press, 1972–1979. [M] ★★★

[D16] T5 Thirring, W. *A Course in Mathematical Physics* (4 vols.). Springer, 1978–1983. [M] ★★

[D17] T3 T5 von Neumann, J. *Mathematical Foundations of Quantum Mechanics.* Princeton University Press, 1955. [M] ★★★

E Primary Research Papers

[E1] T5 Aharonov, Y. & Bohm, D. Significance of electromagnetic potentials in the quantum theory. *Phys. Rev.* **115**, 485 (1959). [P] ★★★

[E2] T5 Aspect, A., Grangier, P. & Roger, G. Experimental tests of Bell's inequalities using time-varying analyzers. *Phys. Rev. Lett.* **49**, 1804 (1982). [P] ★★★

[E3] T5 Bardeen, J., Cooper, L. N. & Schrieffer, J. R. Theory of superconductivity.

Phys. Rev. **108**, 1175 (1957). [P] ★★★

[E4] T4 Becke, A. D. Density-functional thermochemistry. III. *J. Chem. Phys.* **98**, 5648 (1993). [P] ★★

[E5] T3 Bell, J. S. On the Einstein Podolsky Rosen paradox. *Physics* **1**, 195 (1964). [P] ★★★

[E6] T4 Berry, M. V. Quantal phase factors accompanying adiabatic changes. *Proc. R. Soc. Lond. A* **392**, 45 (1984). [P] ★★★

[E7] T3 Born, M. Zur Quantenmechanik der Stoßvorgänge. *Z. Phys.* **37**, 863 (1926). [P] ★★★

[E8] T3 Born, M. & Jordan, P. Zur Quantenmechanik. *Z. Phys.* **34**, 858 (1925). [P] ★★

[E9] T5 Dirac, P. A. M. The quantum theory of the emission and absorption of radiation. *Proc. R. Soc. Lond. A* **114**, 243 (1927). [P] ★★

[E10] T3 Einstein, A., Podolsky, B. & Rosen, N. Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.* **47**, 777 (1935). [P] ★★

[E11] T5 Gleason, A. M. Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.* **6**, 885 (1957). [P] ★★★

[E12] T5 Gorini, V., Kossakowski, A. & Sudarshan, E. C. G. Completely positive dynamical semigroups of N-level systems. *J. Math. Phys.* **17**, 821 (1976). [P] ★★★

[E13] T3 Heisenberg, W. Über quantentheoretische Umdeutung. *Z. Phys.* **33**, 879 (1925). [P] ★★

[E14] T3 Heisenberg, W. Über den anschaulichen Inhalt. *Z. Phys.* **43**, 172 (1927). [P] ★★

[E15] T5 Heisenberg, W. Zur Theorie des Ferromagnetismus. *Z. Phys.* **49**, 619 (1928). [P] ★★

[E16] T4 Hohenberg, P. & Kohn, W. Inhomogeneous electron gas. *Phys. Rev.* **136**, B864 (1964). [P] ★★★

[E17] T5 Keldysh, L. V. Diagram technique for nonequilibrium processes. *Sov. Phys. JETP* **20**, 1018 (1965). [P] ★★★

[E18] T4 Koch, J. et al. Charge-insensitive qubit design. *Phys. Rev. A* **76**, 042319 (2007). [R] ★★

[E19] T4 Kohn, W. & Sham, L. J. Self-consistent equations. *Phys. Rev.* **140**, A1133 (1965). [P] ★★★

[E20] T4 Kubo, R. Statistical-mechanical theory of irreversible processes. *J. Phys. Soc. Japan* **12**, 570 (1957). [P] ★★★

[E21] T5 Lindblad, G. On the generators of quantum dynamical semigroups. *Commun. Math. Phys.* **48**, 119 (1976). [P] ★★★

[E22] T5 Meir, Y. & Wingreen, N. S. Landauer formula for the current. *Phys. Rev. Lett.* **68**, 2512 (1992). [P] ★★

[E23] T4 Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865 (1996). [P] ★★

[E24] T4 Perdew, J. P. & Zunger, A. Self-interaction correction. *Phys. Rev. B* **23**, 5048 (1981). [P] ★★

[E25] T4 Pulay, P. Ab initio calculation of force constants. *Mol. Phys.* **17**, 197 (1969). [P] ★★

[E26] T4 Runge, E. & Gross, E. K. U. Density-functional theory for time-dependent systems. *Phys. Rev. Lett.* **52**, 997 (1984). [P] ★★

[E27] T3 Schrödinger, E. Quantisierung als Eigenwertproblem (Parts I–IV). *Ann. Phys.* (1926). [P] ★★

[E28] T5 Stone, M. H. On one-parameter unitary groups in Hilbert space. *Ann. Math.* **33**, 643 (1932). [P] ★★★

[E29] T3 Wigner, E. P. On unitary representations of the inhomogeneous Lorentz group. *Ann. Math.* **40**, 149 (1939). [P] ★

[E30] T5 Wigner, E. P. The unreasonable effectiveness of mathematics. *Commun. Pure Appl. Math.* **13**, 1 (1960). [P] ★

[E31] T5 Zurek, W. H. Pointer basis of quantum apparatus. *Phys. Rev. D* **24**, 1516 (1981). [P] ★★

F Review Articles & Surveys

[F1] T5 Rotter, S. & Fan, S. Optical cavities during the age of quantum information. *Science* **358**, 1444 (2017). [R] ★

[F2] T5 Zurek, W. H. Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.* **75**, 715 (2003). [R] ★★★

G Historical Collections & Source Books

[G1] T3 Bell, J. S. *Speakable and Unspeakable in Quantum Mechanics* (2nd ed.). Cambridge University Press, 2004. [M] ★★★

[G2] T3 van der Waerden, B. L. (ed.). *Sources of Quantum Mechanics.* Dover, 1967. [M] ★★

[G3] T3 Wheeler, J. A. & Zurek, W. H. (eds.). *Quantum Theory and Measurement.* Princeton University Press, 1983. [M] ★★

[G4] T4 Jammer, M. *The Conceptual Development of Quantum Mechanics.* McGraw-Hill, 1966. [M] ★★

[G5] T4 Mehra, J. & Rechenberg, H. *The Historical Development of Quantum Theory* (6 vols.). Springer, 1982–2001. [M] ★

Tier–Category Quick-Index

Tier		Recommended categories and key items
T1	HS	Cat. A (all); Cat. B: [B8, B19]; no primary papers
T2	BegUG	Cat. B: [B1–B4, B6, B8, B9, B11–B13, B19–B21]; Cat. A as context; Cat. E: [E7] abstract only
T3	AdvUG	Cat. B (all); Cat. G: [G1–G3]; Cat. E: [E5, E7, E8, E10, E13, E14, E27] in full
T4	MSc	Cat. C: [C3, C5, C7–C11, C13, C16, C22–C26]; Cat. D: [D8, D12, D14]; Cat. E: [E1, E6, E7, E11, E16, E19, E20, E23] as primary reading; Cat. G: [G1, G3, G4]
T5	PhD	Cat. D (all); Cat. E (complete); Cat. F (all); Cat. C as reference; current arXiv preprints
Research		Cat. D (all); Cat. E (all); Cat. F (all); current arXiv; journal club